

**NATIONAL INSTITUTE OF ELECTRONICS AND INFORMATION
TECHNOLOGY, CALICUT**



Lab Workshop Manual

**Lab Workshop on FPGA Architecture and Programming using
Verilog HDL**



Contact Point for the Lab

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FPGA Synthesis and Prototyping Lab: 39

Counters

Introduction

The purpose of this experiment is to Synthesis and Check the behavior of the Counters mentioned in Lab 18,19,20,21.

Software tools Requirement

Xilinx Vivado

Synthesis the circuits given in Lab 18, 19, 20, 21.on ARTY A7 FPGA or NEXYS A7 FPGA Kit associated with your Login.

Results should have

Synthesis Report

RTL Schematic

Pin Mapping Report

Screenshot of the Remote Lab-Showing outputs (if any)



Note: You can use Virtual Input /Output to give inputs and View outputs or you may use Chipscope (ILA) for viewing the outputs.